

TEST PLAN

Guru99_Banking_automation

1. INTRODUCTION

This document defines the testing strategy for the Guru99 Banking Application. The goal is to ensure the functional integrity of core modules through automated testing.

- **Core Modules:**
- Login validation
- Customer management
- Account Management
- Deposit
- Withdrawal
- Fund Transfer.

2. SCOPE & FEATURES

Features to be Tested:

- Login (Valid, Invalid, and Empty cases).
- Customer lifecycle (Creation and Validation).
- Account creation and Balance validation.
- Transaction operations: Deposit, Withdrawal, and Fund Transfer.

Features NOT to be Tested:

- Advertisement banners, Header/Footer UI elements.
- Third-party external links, Performance, Security, and Database-level testing.

3. STRATEGY & TOOLS

Approach:

- **Automation:** Selenium WebDriver with TestNG framework.
- **Build/CI:** Maven for dependency management and Jenkins for CI/CD.
- **Reporting:** Extent Reports for detailed HTML execution insights.

Technical Environment:

- **Languages/Tools:** Java, Selenium, TestNG, Maven, Jenkins, GitHub, Chrome/Edge.
- **Hardware:** Windows 10/11, 8GB RAM.

4. EXECUTION CRITERIA

Pass/Fail Metrics:

- Total Test Cases: 12
- Executed: 12
- Passed: 12 | Failed: 0
- **Pass Percentage: 100%**

Suspension & Resumption:

- **Suspension:** Triggered by site unavailability, HTTP 500 errors, driver failure, or Jenkins build failure.
- **Resumption:** Testing resumes once the environment is restored, defects are fixed, and data is verified.

5. RISKS & CONTINGENCIES

- **Site Instability:** Guru99 server-side downtime.
- **Infrastructure:** Browser compatibility, Internet latency, and Jenkins config issues.
- **Mitigation:** Use of **RetryAnalyzer** and robust **WaitUtils** (Explicit Waits) to handle stability.

6. PROJECT DELIVERY & SCHEDULE

Deliverables:

- Automation Framework (GitHub), Test Cases, Extent Reports.
- Jenkins Build Reports, Bug Reports, and Test Summary.

Timeline (7-Day Sprint):

Day	Activity
Day 1	Planning
Day 2	Test Design
Day 3	Framework Setup
Day 4	Script Development
Day 5	Execution
Day 6-7	Retesting & Cleanup

7. ROLES & APPROVALS

- **SDET Trainee:** Bharath kumar
- (Test Design, Automation, Execution, Bug Fixing, Jenkins Pipeline).
- **Trainer:** Ms. Vaishali Ahire.

- **Project Name:** Guru99_ Banking_ Automation